

Computer Modeling & Simulation Problems

Professional football coach has six running backs on his squad. He wants to evaluate how injuries might affect his stocks of running backs. A minor injury causes a player to be removed from the game and miss only the next game. A major injury puts the player out of action for the rest of the season. The probability of a major in a game is 0.05. There is at most one major injury per game. The probability of minor injury per game is:

Number of injury	Probability
0	0.2
1	0.5
2	0.22
3	0.05
4	0.025
5	0.005

Injuries seem to happen in a completely random pattern over the season. A season is 10 games. Using the following random number, simulate the fluctuating in the coach's stock of running backs over the season. Assume that he hires no additional running backs during the season:-

R.N:

**044 392 898 615 986 959 558 353 577 866 305 813 024 189 878 023
285 442 862 848 060 131 963 874 805 105 452 .**

Solution:

Number of injury	Probability
0	0.200 000 → 199
1	0.500 200 → 699
2	0.220 700 → 919
3	0.050 920 → 969
4	0.025 970 → 994
5	0.005 995 → 999

- Note we have three digit random numbers that means numbers range 000 → 999 (1000 integer numbers)

- Probability of **major** 0.05 (000 → 049) take 50 number from 1000

Then **probability of not having major is** the remaining numbers 050 → 999

- There is at most one major injury per game.

Game #	No. of players	R.N major	YES/NO	R.N minor	No. of minor	STOCK
1	6	044	YES	392	1	4
2	4	898	NO	615	1	3
3	4	986	NO	959	3	1
4	2	558	NO	353	1	1
5	4	577	NO	866	2	2
6	3	305	NO	813	2	1
7	3	024	YES	189	0	2
8	4	878	NO	023	0	4
9	4	285	NO	442	1	3
10	3	862	NO	848	2	1